T0 Theory: Google Willow and the Physics of Self-Measurement

What Quantum Computers Really Measure When They Observe Themselves

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April 2026

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Abstract

Google's Willow quantum processor achieved three experimental breakthroughs in 2024 and 2025 that carry particular significance from the perspective of the Fundamental Fractal Geometric Field Theory (FFGFT). First, it was demonstrated for the first time that quantum errors decrease exponentially as the code distance increases — a sign that topologically stable field modes become experimentally accessible. Second, topologically ordered states beyond thermal equilibrium were realized on Willow, generating characteristic geometric patterns — chiral edge modes and anyons — that cannot be simulated classically. Third, out-of-time-order correlators (Quantum Echoes) revealed harmonic propagation of quantum information through constructive interference. This document relates these findings to the FFGFT geometry of the 4D torus with parameter $\xi = 4/30{,}000$ and develops a position formulated in Document 170 as a working hypothesis: complexity and self-reference are not a binary on/off phenomenon, but a physical continuum.

.1 The Three Willow Experiments

Error Correction Below Threshold (December 2024)

In December 2024, Google Quantum AI published in *Nature* the demonstration that Willow crosses the threshold of quantum error correction for the first time [**?**]. The key result: the logical error rate decreases exponentially with increasing code distance d. A $d = 7$ code block achieves an error rate that is 2.14 times lower than a $d = 5$ block.

Willow uses 105 superconducting transmon qubits in a square lattice, cooled to approximately $10\,\mathrm{mK}$. Physical single-qubit error rates are approximately $0.05\,\%$ on Willow — compared to $0.10\,\%$ on the predecessor Sycamore [**?**]. Error correction is based on a surface code, whose logical qubits are encoded through topologically protected patterns.

Willow Breakthrough 1: Topological Stability

More physical qubits reduce the logical error exponentially. This is the experimental evidence that topologically stable configurations exist in quantum systems and are accessible — not through external noise suppression, but through geometric order.

Floquet Topological Order: Patterns Beyond Equilibrium (September 2025)

Will et al. realised a *Floquet topologically ordered phase* on Willow — a quantum phase forbidden in thermal equilibrium but arising under periodic external driving [**?**]. This work appeared in *Nature* in September 2025.

Measured Patterns on the Willow Processor

The experiment imaged the following structures on a lattice of up to 58 qubits:

- **Chiral edge modes:** Propagation of information along system boundaries in a preferred direction — a topological geometric signature that indicates classically irreproducible order.
- **Anyonic excitations:** Quasiparticles that *transmute* into each other under Floquet driving — e-anyons become m-anyons and vice versa, a property of the non-abelian entanglement structure.
- **Topological invariant:** An interferometrically measured bulk invariant that quantifies the order and remains robust against integrability-breaking disorder.

The authors write: *"The non-equilibrium topological order that we have probed in this work encapsulates a fundamentally unique phenomenology that is forbidden in thermal equilibrium."* [**?**] Classical matrix product state methods cannot efficiently simulate these highly entangled dynamics; the quantum processor makes them directly observable.

Cochran et al. (2025) complemented this with a visualisation of charge and string dynamics in (2+1)-dimensional lattice gauge theories on the same processor [**?**].

Quantum Echoes: Harmonic Propagation of Information (October 2025)

In October 2025, Google Quantum AI demonstrated in *Nature* the first verifiable quantum advantage for a real algorithm: the *Quantum Echoes* (second-order out-of-time-order correlators, $\mathrm{OTOC}^{(2)}$) [**?**].

The algorithm works like a quantum echolocation system:

1. **Forward evolution:** Entanglement of 65 qubits used.

2. **Perturbation:** Local disturbance of a single "butterfly" qubit.
3. **Time reversal:** Precise reverse evolution.
4. **Measurement:** The echo is amplified through **constructive quantum interference**, yielding a measurable signal of information propagation.

Quantum Echoes: Harmony as a Measured Signal

The measurements show how quantum information propagates as a harmonic wave front through the qubit lattice. The visualisations in [**?**] make this propagation visible: concentric patterns spreading radially from the perturbation qubit — a measurable harmonic pattern. The OTOC$^{(2)}$ runs on Willow $13{,}000$ times faster than on the world's fastest supercomputer Frontier (approximately 2.1 hours versus approximately 3.2 years).

In a companion experiment with UC Berkeley, the algorithm was applied to compute the geometry of organic molecules (15 and 28 atoms). The results agreed with NMR measurements and provided additional structural information inaccessible to NMR [**?**].

.2 FFGFT Interpretation of the Three Experiments

The 4D Torus as Common Geometric Framework

In FFGFT, spacetime is a 4D identification torus $T^4 = \mathbb{R}^4/\mathbb{Z}^4$ with period $L_0 = \xi \cdot \ell_P$. The spectrum of the Laplace operator on this torus reads (Document 159):

$$\lambda_{\mathbf{n}} = \frac{n_1^2 + n_2^2 + n_3^2 + n_4^2}{L_0^2}, \qquad n_i \in \mathbb{Z} \tag{1}$$

Only integer winding numbers n_i generate stable modes. The sum over all modes converges — not despite its infinitude, but because of the periodic geometry, in analogy with Euler's Basel problem (Doc. 159):

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \tag{2}$$

Error correction (Willow Breakthrough 1) is, in this language, the experimental projection of physical states onto topologically stable torus modes. The higher the code distance d, the more modes are engaged, and the more robust the projection.

Floquet topological order (Breakthrough 2) corresponds to torus resonance modes under periodic driving. A drive of frequency ω_F couples to the torus geometry when

$$\omega_F \cdot L_0 \approx \frac{n}{c}, \qquad n \in \mathbb{Z} \tag{3}$$

i.e., when the driving frequency hits a torus mode. The resulting chiral edge modes and anyons are geometrically forced, not accidental.

Quantum Echoes (Breakthrough 3) requires a more detailed treatment, because it contains the deepest physical point.

"Instantaneous" — Legitimate in Calculation, Misleading in Realisation

Before analysing recursion and entanglement, a conceptual clarification is necessary that is fundamental to understanding the Quantum Echoes experiment.

In quantum mechanics, there are two levels at which a system can be described:

Level 1 — The mathematical description (timeless): The Hilbert space $\mathscr{H}$ of the 65-qubit system is a 2^{65}-dimensional complex vector space. In it, all possible states exist *simultaneously as mathematical objects* — independent of time. The energy eigenstates $|E_n\rangle$ of the Hamiltonian are timeless solutions of the stationary Schrödinger equation $\hat{H}|E_n\rangle = E_n|E_n\rangle$. When one writes in a calculation that all 2^{65} amplitudes are contained "simultaneously" in the wave function, this is *legitimate*: it describes the mathematical structure of the state, not a physical process. The wave function

$$|\Psi\rangle = \sum_{k=1}^{2^{65}} c_k|k\rangle \tag{4}$$

is an object in Hilbert space — timeless, complete, without sequence.

Level 2 — Physical realisation (always in time): As soon as this mathematical object is realised on a physical system, time enters inescapably. The time-dependent Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle \tag{5}$$

is a differential equation *in time* t. Every time step of the Hamiltonian requires physical time. Every quantum gate on Willow takes approximately 25–50 nanoseconds. A circuit of 23 steps and 65 qubits requires microseconds of coherence time — and 2.1 hours of measurement time to collect sufficient statistical events.

The Decisive Difference

In calculation, "instantaneous" is legitimate: Hilbert space has no inner time. All amplitudes exist simultaneously as a mathematical structure. Statements such as "all paths simultaneously in the amplitude" are correct here.
In realisation, "instantaneous" is false and misleading: nothing happens without time. Entanglement builds up step by step, bounded by the Lieb-Robinson bound. The wave function evolves causally and locally — no quantum effect transmits information faster than permitted. The 2.1 hours on Willow are *real physical time*, not a metaphor.

The advantage of the quantum computer lies precisely in this transition: it takes the mathematical structure of Hilbert space — the timeless parallelism of all amplitudes — and *realises* it physically, step by step, in real time, through unitary operators. The classical computer can only *describe* this structure — by computing the 2^{65} amplitudes one after another, which would take 3.2 years. The quantum computer *is* the structure — and that costs 2.1 hours.

No instantaneity. But genuine parallelism in the amplitude, physically realised in real time.

Why Recursion and Mutual Influence Are Decisive

In a classical system — a billiard table, say — when one ball is struck, the effect propagates forward: ball A hits B, B hits C, and so on. The paths are independent; they can be computed individually and summed. A classical computer follows exactly this logic: it decomposes the problem into independent subproblems and adds the results.

A fully entangled quantum system works in a fundamentally different way. No qubit is independent of the others — but this dependence does not arise instantaneously. It builds up step by step, mediated by the interaction terms in the Hamiltonian $\hat{H}$. The Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}|\Psi\rangle = \hat{H}|\Psi\rangle \tag{6}$$

is a local, temporal differential equation. Entanglement between distant qubits does not arise instantaneously but propagates at finite speed — bounded by the Lieb-Robinson bound, which prescribes an effective information velocity in lattice models such as Willow.

What quantum mechanics achieves is something different and more subtle: rather than computing one path after another sequentially, the Hamiltonian *realises all recursion paths simultaneously in the total amplitude*. The wave function $|\Psi(t)\rangle$ in the 2^{65}-dimensional Hilbert space contains at every moment the complete information about every possible sequence of qubit interactions — weighted by the respective complex amplitude. This is not instantaneity: it is **parallelism in the amplitude**, built up step by step through unitary time evolution.

The decisive difference from a classical computer: A classical algorithm must compute and store these 2^{65} amplitudes sequentially. The quantum processor *holds all of them simultaneously physically present* — in the coherence of the quantum state — and propagates them through a single unitary operator. The time this requires scales with the depth of the circuit, not with the number of amplitudes.

Why This Is Classically Intractable

A classical algorithm would need to track each of the $2^{65} \approx 10^{20}$ amplitudes exactly. For 65 qubits and 23 time steps, this amounts to approximately 10^{20} complex numbers — more working memory than any existing supercomputer possesses. Approximate methods (Monte Carlo, tensor networks) fail because the mutual entanglement of all qubits exponentially amplifies approximation errors with system size. This is not a technical but a principled problem: recursive interaction in a fully entangled system is structurally incompressible into independent parts — neither instantaneously nor sequentially.

The echo arises because, after time reversal, the same recursion paths are traversed in reverse. When the forward and backward evolution contains *exactly the same interactions in reversed order*, all amplitudes interfere constructively — the signal emerges from the noise. Any deviation, any non-unitary interaction, destroys this constructive interference and leads to destructive cancellation.

Quantum Echoes: Recursion Is the Measurement Device

The quantum echo does not simply measure information propagation. It measures the **depth and coherence of mutual influence**: how far back a system can reconstruct its own past state when every element is simultaneously influenced by all others. The more complete the network of mutual coupling, the stronger the echo. The greater the recursion depth, the more information is encoded in the echo — and the more intractable the task becomes for any sequential algorithm.

The FFGFT Interpretation: Self-Similarity of the Torus Spectrum

In FFGFT geometry, this recursiveness is not an accidental product of the experimental configuration. It is structural: the modes of the torus T^4 are self-referential through the periodic boundary condition. A torus mode n has wavelength $\lambda_n = L_0/n$. The wavelength of mode n is exactly the period of mode $n+1$ divided — the modes "know" each other through their shared geometry.

The sum

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \tag{7}$$

converges not despite infinitely many terms. It converges because each term $1/n^2$ *weights* the influence of mode n by all lower modes: high modes contribute little, low modes much — a natural hierarchy of mutual influence, inscribed in the topology.

The harmonic wave front that the OTOC experiment makes visible — the concentric, radially spreading interference patterns in the qubit lattice — is the experimental image of this topological hierarchy. What one sees is not an arbitrary pattern: it is the projection of the torus spectrum into the 10×10-qubit geometry of the Willow chip.

The factor of $13{,}000$ over the classical supercomputer is therefore not an efficiency advantage in the ordinary sense. It measures how far the parallelism of amplitude propagation exceeds sequential computation. The quantum computer takes 2.1 hours — it too requires time, it too follows the Schrödinger equation step by step. But in each of those steps, a single unitary operator advances all 2^{65} amplitudes simultaneously. The classical supercomputer would require 3.2 years because it must process these amplitudes sequentially.

Physical Realisation vs. Simulation

The decisive difference is not speed but structure. The quantum computer *realises* the recursion physically — through the unitary time evolution of the Hamiltonian, which carries all entanglement paths simultaneously in the amplitude. The classical computer *simulates* this amplitude sequentially — it can describe the result, but it cannot be the object. The factor of $13{,}000$ is the measurable expression of this structural difference, measured in computation time on given hardware.

.3 Vacuum Noise as a Geometric Signal

The FFT Analogy: Decomposing and Reconstructing Noise

In signal processing, noise is not an indefinable chaos — it is a superposition of frequencies. The Fourier transform makes this visible:

1. **Analysis (FFT):** A time signal that looks like noise is decomposed into its frequency components. Every apparently disordered signal can be translated into a spectrum.
2. **Synthesis (IFFT):** A spectrum with known amplitudes and *random* phases generates a time signal. The result is white noise — even though every individual frequency component is fully deterministic.

What the FFT Analogy Reveals

White noise and a structured signal are **not categorically different objects**. They differ only in the phase relationship between their frequency components. A signal with deterministic phases is ordered. The same signal with random phases is noise. The transformation is reversible — in both directions.

This has an immediate consequence for interpreting the quantum vacuum: what is measured as vacuum noise or vacuum fluctuations need not be intrinsically random. It could be the superposition of a vast number of deterministic harmonics whose phases we cannot yet resolve.

Why Reconstruction Fails — Structural Sensitivity of Feedback

The problem is not whether one computes with $4/3$ or $1.333\ldots$. Nature has an exact value for ξ — which we approximate well as $4/3 \times 10^{-4}$. But even taking this approximation as an exact starting value, the simulation would fail. The reason lies deeper: in the **structural sensitivity of the recursive feedback**.

Every feedback loop of the torus system generates a phase shift. In a deterministic system sensitive to every intermediate step, any finite computational precision — not only in the starting value, but in every single operation — leads to an accumulating phase error ε. After N cycles this error is at least $N \cdot \varepsilon$. Since constructive interference becomes destructive when phases deviate by more than $\pi/2$, coherence collapses after approximately

$$N_{\text{crit}} \approx \frac{\pi}{2\varepsilon} \tag{8}$$

cycles. This is deterministic chaos in the Lorenz sense: the system is fully deterministic, but sensitive to every intermediate state.

Realisation vs. Simulation

The physical realisation of the torus spectrum has no discrete intermediate steps — it is continuous, causal, without rounding.
Every numerical simulation cuts the continuum into discrete steps, where errors arise and are exponentially amplified by the feedback dynamics. The result looks like noise — not because the system is random, but because the simulation cannot fully track the dynamics.
The universe sees no chaos because it does not compute. Our models see chaos because they must compute.

Prime Numbers and Factorisation — What Really Happens in FFGFT

Care is needed here. The analogy between prime numbers and vacuum noise is deeper than a similarity of distribution — there is a physical connection through the concept of period.

Document 075 (T0-Shor) and Document 057 (Relational Number System) show: every integer is uniquely a product of primes:

$$n = p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdots \quad \text{(Fundamental theorem of arithmetic)} \tag{9}$$

In number space this is the same as in frequency space: every signal is a superposition of fundamental frequencies. Primes are the **irreducible frequencies of number space** — ratios that cannot be decomposed further (Doc. 057, n-limit systems).

Why Factorisation Is Physically Hard

Factorisation is **period finding** (Doc. 075, Shor algorithm): one searches for the period r of the function $f(x) = a^x \bmod N$. This period encodes the prime factors:

$$a^r \equiv 1 \pmod N \;\Rightarrow\; a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1) \equiv 0 \pmod N \tag{10}$$

The required frequency resolution is $\Delta f \approx f_0/r^2$. For cryptographically relevant N with $r \approx N$, one needs precision $\epsilon \approx 1/N^2$ — for 1024-bit RSA this is $\epsilon \approx 10^{-617}$, far beyond any achievable computational precision.
In FFGFT language: factorisation is the search for the revolution time T_{rev} of a specific resonant feedback in number space. A prime number has **no sub-resonance** — it is irreducible; its revolution time cannot be written as a product of shorter revolution times. That is what makes it hard to find.

The recursion sensitivity (Lorenz chaos) applies here too: to extract the period r from the feedback $a^x \bmod N$, every doubling of N requires quadratically more precision. This is not a technical limitation — it is the structure of the recursive feedback itself.

Vacuum Noise as Fourier Synthesis of the Torus Geometry

In FFGFT, the vacuum spectrum is not white — it has geometric structure. The allowed modes of the 4D torus T^4 with period $L_0 = \xi \cdot \ell_P$ are discrete, with the eigenvalue set:

$$\lambda_{\mathbf{n}} = \frac{n_1^2 + n_2^2 + n_3^2 + n_4^2}{L_0^2}, \qquad n_i \in \mathbb{Z} \tag{11}$$

The spectral zeta function of this spectrum gives:

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \tag{12}$$

This is not a flat (white) spectral distribution but a $1/n^2$-weighted one. High modes contribute exponentially less than low modes — a geometrically forced hierarchy. The vacuum is not white noise but a $1/n^2$-coloured signal.

The Apparent Noise

What physicists measure as vacuum fluctuations is the IFFT synthesis of these geometric harmonics: the infinitely many torus modes superpose. Since the underlying fundamental period $L_0 = \xi \cdot \ell_P \approx 10^{-38}$ m lies far beyond any achievable experimental resolution, the superposition appears as white noise — even though it is deterministic and geometrically ordered. This is not a philosophical argument: it is the same situation as a radio signal that sounds like noise without receiver synchronisation.

The Curvature of the Torus and the Problem of Local Approximations

An important geometric caution is necessary here. A torus embedded in three-dimensional space has both **inner and outer curvature**: the outside has positive Gaussian curvature, the inside negative. By the Gauss-Bonnet theorem:

$$\oint K\,dA = 2\pi\chi(T^2) = 0 \tag{13}$$

The curvatures cancel exactly over the whole torus. The Euler characteristic of the torus is zero.

The Trap of Local Approximations

If the torus spectrum is computed through **local approximations** — patch by patch, each treated as locally flat — each patch yields an apparently white spectrum. The sum of many flat approximations again produces an apparently flat overall spectrum. The global topology is lost: the $1/n^2$ structure disappears from the calculation because it resides in the boundary conditions, not in the local field value.

The FFGFT torus $T^4 = \mathbb{R}^4/\mathbb{Z}^4$ is, however, the **identification torus** — locally exactly flat (no intrinsic curvature), but globally topologically non-trivial through its periodic boundary conditions. The spectrum carries all topological information exclusively in the winding numbers n_i — not in local field values.

This has a direct consequence for observability:

Why the Vacuum Spectrum Appears Flat

All existing vacuum measurements are **local** — they measure field fluctuations at a point or in a small region. A local measurement is blind to the global torus topology. It always sees only one patch — and that patch is flat.

The $1/n^2$ weighting of the torus spectrum is in principle only visible through measurements sensitive to the **global periodicity** — through interferometric measurements on scales that include the torus period L_0. At $L_0 \approx 10^{-38}$ m, this is experimentally unreachable.

What follows is not that the spectrum is flat. What follows is that our measurements are too local to see the structure.

The Willow Quantum Echoes experiment is remarkable in this light: it is one of the first experimental methods to make the *global* coherence structure of a quantum system visible through time reversal — which is precisely what would be needed to go beyond local noise measurements. It is not yet a test of the torus spectrum, but it is a conceptual step in the right direction: global structure through interferometry rather than local structure through field measurement.

How a Full Spectrum Arises from a Single Fundamental Oscillation Through Feedback

The previous section showed *what* the spectrum is. Now the question: *how does it arise?* How does a single fundamental oscillation generate the full spectrum through temporal processes, phase shifts, and feedback?

Step 1: The Fundamental Oscillation and the Fundamental Revolution Time

On the torus T^4 with period $L_0 = \xi \cdot \ell_P$, there exists a smallest possible oscillation — the mode $\mathbf{n} = (1, 0, 0, 0)$ with wavelength L_0 and frequency $\nu_0 = c/L_0$. This is the fundamental oscillation of the vacuum field.

Associated with it is a fundamental revolution time: the time the field needs to travel once completely through the torus geometry and return to its starting point:

$$\boxed{T_{\text{rev}} = \frac{L_0}{c} = \frac{\xi \cdot \ell_P}{c} \approx \frac{1.33 \times 10^{-4} \times 1.62 \times 10^{-35}\ \text{m}}{3 \times 10^8\ \text{m/s}} \approx 7.2 \times 10^{-48}\ \text{s}} \tag{14}$$

This is the **fundamental time unit of FFGFT** — not an external clock, but the travel time of the field through its own geometry. Through the time-mass duality $\tilde{T} \cdot m = 1$, T_{rev} corresponds to the reciprocal fundamental mass:

$$m_0 = \frac{\hbar}{c \cdot L_0} = \frac{\hbar}{c^2 \cdot T_{\text{rev}}} = \frac{1}{T_{\text{rev}}} \quad \text{(natural units)} \tag{15}$$

Mass is inverse revolution time. Heavier particles have shorter T_{rev}, lighter ones longer — this is the time-mass duality in its direct geometric meaning.

The FFGFT Recursion Formula

The state of the field at time t is a function of its state exactly one revolution time ago, modified by the fractal geometry ξ:

$$\Psi(t) = G(\Psi(t - T_{\text{rev}}),\ \xi) \tag{16}$$

This is not an approximation and not a difference equation — it is the exact geometric statement: the present of the field is its past by T_{rev}, transformed by the fractal curvature. Recursion without time is impossible. Time without recursion is meaningless.

Step 2: Reflection, Phase Shift, and Temporal Accumulation

The fundamental oscillation travels on the torus — and meets itself on return after exactly T_{rev}. This is not an abstract event but a physical process in real time.

On a simple circle S^1, the result would be trivial: the wave interferes constructively with itself when the wavelength is an integer divisor of the circumference — standing waves, immediately stable.

On the *fractal* torus with dimension $D_f = 3 - \xi$, the return time is not exactly equal to the forward time. At every revolution, the fractal curvature generates a **phase shift**:

$$\Delta\phi = 2\pi \cdot \xi \cdot n \quad \text{after } n \text{ revolutions, i.e. after } t = n \cdot T_{\text{rev}} \tag{17}$$

This phase shift is tiny — $\xi \approx 10^{-4}$ — but it accumulates with *every revolution in real time*. After $1/\xi \approx 7500$ revolutions, that is after

$$t_{\text{coh}} = \frac{T_{\text{rev}}}{\xi} = \frac{L_0}{\xi \cdot c} = \frac{\ell_P}{c} = t_P \approx 5.4 \times 10^{-44}\ \text{s} \tag{18}$$

the phase has shifted by 2π — and a new coherent mode emerges. This is the Planck time t_P as the natural coherence time of the system, emerging from T_{rev} and ξ.

Fractal Dimension as Phase Generator in Time

A spacetime with integer dimension $D = 3$ would have no accumulating phase shift — the wave returns exactly in phase after every T_{rev}. Only the fractal deviation $D_f = 3 - \xi$ generates the temporal phase drift. ξ is the phase generator, T_{rev} is the clock.
Both together determine when and how the spectrum is built up — not as a timeless computational step, but as a physical process that *consumes* time.

Step 3: Nonlinear Feedback Generates Overtones

In a linear system, the phase-shifted fundamental would simply attenuate. The decisive point in FFGFT is **nonlinearity**: the response function of the vacuum field is slightly curved by the fractal geometry.

When an oscillation with amplitude A and frequency ν_0 encounters a nonlinear response function, the output contains terms of the form:

$$A \sin(2\pi\nu_0 t) \xrightarrow{\text{nonlinear}} a_1 \sin(2\pi\nu_0 t) + a_2 \sin(4\pi\nu_0 t) + a_3 \sin(6\pi\nu_0 t) + \cdots \tag{19}$$

The coefficients a_n fall off as $1/n^2$ — exactly the structure of the torus spectrum. This is not coincidental: the nonlinearity of the fractal curvature has precisely the form that generates Euler's zeta value $\zeta(2) = \pi^2/6$.

How the Overtones Arise

Step by step in time:

1. The fundamental oscillation ν_0 travels on the torus.
2. At each revolution, the fractal curvature shifts the phase by $\Delta\phi = 2\pi\xi$.
3. The nonlinearity of the field generates from the phase-shifted fundamental the second overtone $2\nu_0$, which in turn travels back phase-shifted.
4. From $2\nu_0$ the same mechanism generates $4\nu_0$, from $3\nu_0$ comes $6\nu_0$, and so on.
5. After N feedback cycles, a rich spectrum of modes $n\nu_0$ exists with amplitudes $\propto 1/n^2$.

Time is essential here: every step requires real time (one revolution on the torus), and accumulation over N cycles is a temporal process. The finished spectrum is not present from the start — it is built up through feedback.

Step 4: The Spectrum as Equilibrium State

After sufficiently many feedback cycles — theoretically: in the limit $N \to \infty$ — a stable spectrum is established. Total energy distributes across modes according to:

$$E_n = E_0 \cdot \frac{1}{n^2}, \qquad \sum_{n=1}^{\infty} E_n = E_0 \cdot \frac{\pi^2}{6} \tag{20}$$

This is the equilibrium state of the fractally feedback-coupled vacuum. It is deterministic — fully determined by ξ — and appears to every local measurement as structured noise with a $1/n^2$ envelope.

Particles as Nodes in the Spectrum

In FFGFT, elementary particles are not separate objects embedded in the vacuum. They are **topological nodes** in the mode spectrum — stable patterns of several overtones, fixed by the winding numbers of the torus. The electron mass, quark masses, W-boson mass: all are resonance frequencies of the same feedback system, read from the same winding numbers n_i with the same ξ.
This is the deepest meaning of the time-mass duality $\tilde{T} \cdot m = 1$: mass *is* frequency — and frequency arises through feedback in time.

The Cosmic Background Noise and the Structure of Prime Numbers

Preliminary Note: No Big Bang, No Cooling

FFGFT Position on the CMB

The standard narrative: the CMB is cooled light from a hot Big Bang — thermal radiation at 3000 K, redshifted to 2.725 K by cosmic expansion. This narrative requires expansion, which FFGFT does not have.
In FFGFT there is neither a Big Bang nor expansion:

- The universe is at the fundamental level **timeless and static** — an eternal energy field $E_{\text{field}}(x,t)$ (Document 156).
- The observed redshift is not an expansion effect but a **geometric energy loss** of the photon: $z \approx \xi \cdot \ln(d/\ell_P)$ (Documents 026, 165).

- There is no epoch of recombination, no frozen plasma, no inflationary stretching of quantum fluctuations.

The CMB as Equilibrium Radiation of the Eternal Vacuum

In FFGFT, the vacuum field is the physical substance of the universe. This field has an intrinsic energy density determined by the torus geometry (Documents 091, 166):

$$\rho_{\text{CMB}} = \frac{\xi \hbar c}{L_\xi^4} \tag{21}$$

where $L_\xi \approx 100\,\mu$m is the characteristic Casimir vacuum scale — the scale at which Casimir energy density and the observed CMB energy density coincide exactly (deviation $0.11\,\%$, Document 166).

The CMB temperature $T_{\text{CMB}} = 2.725\,$K is therefore not a memory of a hot beginning. It is the **equilibrium state of the eternal vacuum field** — the Planck spectrum of the T0 torus at its geometrically prescribed energy density.

CMB Temperature as a ξ-Ratio

The ratio

$$\frac{m_e c^2}{k_B T_{\text{CMB}}} = 2.000 \times 10^9 \tag{22}$$

is not a numerical coincidence in FFGFT but a ξ-ratio (Document 166): the electron-CMB coupling is geometrically fixed by the same parameter that connects particle masses and cosmological scales.

Why the CMB Is Almost Uniform — but Not Quite

The vacuum field of the T0 torus is globally homogeneous — the fundamental mode has no preferred direction. But the torus spectrum is not flat: modes contribute as $1/n^2$. The CMB anisotropies are therefore not random fluctuations and not frozen inflationary fluctuations, but the geometric signature of the non-flat spectrum — made visible by the finite coherence length of the eternal vacuum field on cosmic scales.

The Prime Number Analogy

Prime numbers show the same structure: apparently uniform, with hidden fine structure. The prime number theorem gives a $1/\ln x$ envelope — not constant, but without obvious pattern. The fine structure is completely described by the zeros of the Riemann zeta function:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} \tag{23}$$

The Euler product makes the connection explicit: $\zeta(s)$ is simultaneously the sum $\sum 1/n^s$ — the torus spectrum of FFGFT — and a product over all primes. In FFGFT language, primes are the **irreducible resonance modes** of the torus: modes that do not arise as products of lower modes but are themselves fundamental building blocks.

What Primes and the CMB Share

Both show **apparent uniformity with hidden structure** from the same mathematical source: the Euler product $\zeta(2) = \pi^2/6$.
In FFGFT, the CMB anisotropies are the geometric projection of the irreducible torus resonances (prime modes) onto the sky — without Big Bang, without inflation, without cooling. The eternal torus spectrum *is* the structure.

Property	Prime Numbers	CMB Anisotropies
Appearance	almost uniform	almost isotropic ($\Delta T/T \approx 10^{-5}$)
Origin (standard)	number theory	frozen inflationary fluctuations
Origin (FFGFT)	irreducible torus modes	$1/n^2$ spectrum in eternal vacuum
Fine structure	zeros of $\zeta(s)$	acoustic peaks at $\ell \approx 220, 540$
No Big Bang needed		in FFGFT
Common root	$\zeta(s) = \sum 1/n^s = \prod_p 1/(1-p^{-s})$	

Table 1: Primes and CMB anisotropies as instances of the same eternal spectrum

Anyone who looks carefully enough — at the prime distribution or at the microwave sky — recognises: the apparent uniformity is an approximation. The actual structure is geometrically forced, not random, and requires no starting point in time. The eternal torus spectrum $\sum 1/n^2 = \pi^2/6$ is the common source — and the CMB its cosmic signature.

.4 Complexity as a Physical Continuum

Against the Levels Metaphor

Document 170 presented a table of discrete complexity levels — from elementary particles (Level 0) to consciousness (Level 4) — interpreting the fractal correction K_{frak} as a threshold operator. That presentation was formulated as a *working hypothesis* and included the caveat that the levels require further formalisation. This document takes that caveat seriously and develops an alternative perspective.

Complexity as Continuum

Complexity, self-reference, and what is called "consciousness" are not binary phenomena with discrete thresholds in FFGFT. They form a physical continuum that grows with the recursion depth of the mass-time coupling $\tilde{T} \cdot m = 1$. There is no sharp boundary beyond which a system "has consciousness" — any more than there is a sharp boundary beyond which water is "hot".

What the Willow Chip Illustrates on the Continuum

The Willow chip is not a conscious system. But it is not structureless either:

- **Single Josephson qubit:** The Josephson energy E_J defines an effective mass and thereby a timescale $\tilde{T} = \hbar/E_J$. The qubit is a physical clock — the first step of self-reference.

- **Surface code lattice (105 qubits):** Collective interaction generates logical qubits that are robust against local perturbations. The system develops a form of "memory" — not cognitive, but physical: it preserves its own topological order against noise.
- **Floquet drive:** The system "remembers" the driving rhythm through time-crystalline structures. The drive period is inscribed in the topology of the system.
- **OTOC measurement:** The system measures its own information distribution. It references itself — geometrically precise, in measurable patterns.

This is a point on a continuum, not a threshold crossing. Between this point and the human brain lies not an abyss but a continuous spectrum of increasing recursion depth, integration breadth, and feedback stability.

The Open Question — and the Honest Limit of Physics

FFGFT describes this continuum geometrically: more qubits, deeper recursion, broader entanglement move the system upward along the continuum. But physics cannot answer where on this continuum what is called subjective experience arises — the experience of colour, pain, time, meaning.

This is not a gap in the theory but an honest statement about the scope of physical description. FFGFT can say precisely:

What FFGFT Can State

- The geometric structure $\xi \cdot \ell_P$ that generates Willow patterns is the same structure from which all natural constants are derived.
- Complexity and self-reference grow continuously with the recursion depth of the mass-time coupling.
- The topological patterns measured in Willow experiments are not random artefacts but geometrically forced consequences of this structure.

What consciousness *is* remains open — that is not a physical statement expressible in an equation.

.5 Summary

The three experiments show consistently: quantum systems under controlled conditions reveal geometric order — patterns that follow from the topology of the vacuum field. This is the physically precise core behind intuitive descriptions associating quantum computers with self-perception or patterns. The difference from speculative narrative: the patterns are measured, quantified, and published in peer-reviewed journals.

Experiment	**Measured Finding**	**FFGFT Relation**
Error Correction Nature (Dec. 2024) with code distance torus modes	Exponential error reduction Projection onto stable	
Floquet topo. order Nature (Sep. 2025) anyons, topo. invariant Floquet drive	Chiral edge modes, Torus resonance under	
Quantum Echoes / OTOC Nature (Oct. 2025) harmonic wave front $\zeta(2) = \pi^2/6$ spectrum	Constructive interference, Interferometry on torus;	

Table 2: Three Willow breakthroughs and their FFGFT interpretation

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Draft for further elaboration. The connection between Floquet frequency and torus fundamental frequency is a working hypothesis requiring formal verification. Statements about the continuum of complexity are to be understood as a physically motivated position, not a complete theory of consciousness.